Forecasting Demand for National Research Supercomputing

Australian Academy of Science | Australian HPC Demand to 2030

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Chapter 1

Introduction and Objectives

1.1 Project Background and Motivation

High-performance computing (HPC) is an important component of Australia's research infrastructure, supporting computationally intensive research across a wide range of scientific fields (Australian Academy of Science 2024). Australia has two national Tier-1 HPC facilities accessible to researchers: the National Computational Infrastructure (NCI) and the Pawsey Supercomputing Research Centre (National Computational Infrastructure, n.d.; Pawsey Supercomputing Research Centre, n.d.).

As research becomes increasingly data- and compute-intensive, understanding how demand for HPC resources is changing is important for planning Australia's future national research infrastructure (Australian Academy of Science 2024).

The purpose of this project, undertaken with the Australian Academy of Science, is to analyse historical Australian research computing demand and develop evidence-based projections of future HPC demand to 2030. The analysis is intended to support national policy discussions relating to public compute investment and future research infrastructure planning (Australian Academy of Science 2026).

The project combines data engineering, descriptive analysis and forecasting using publicly available NCI and Pawsey data, particularly information from the National Computational Merit Allocation Scheme (NCMAS). Additional indicators are considered where sufficiently consistent and relevant data are available (Australian Academy of Science 2026).

1.2 The Forecasting Problem

One of the main challenges in forecasting national research computing demand is that observed HPC allocation does not necessarily represent the full level of research demand.

National allocation schemes operate under finite capacity. As a result, allocated compute represents the amount of resources awarded rather than necessarily the amount researchers would use if additional capacity were available. Requested compute provides additional evidence of demand, but it may also understate underlying need where researchers constrain their requests based on expected availability or where potential users do not apply (Australian Academy of Science 2026).

The analysis therefore distinguishes between requested compute, allocated compute, the gap between requested and allocated compute, and broader underlying or latent demand, which cannot be observed directly from the available allocation data.

A further challenge is the comparability of historical HPC information. Reporting practices, hardware systems, compute units and service-unit accounting have changed over time. CPU and GPU resources are also not

directly interchangeable, while changes in Field of Research (FoR) classifications affect comparisons of research allocation across years.

These limitations are particularly important because the available annual time series is relatively short. The analysis therefore documents where historical values are directly reported, derived, estimated or based on assumptions. Forecasting results are interpreted alongside these limitations rather than as precise measurements of unconstrained demand.

1.3 Project Objectives

The primary objective is to develop evidence-based projections of Australian national research computing demand to 2030, while accounting for limitations in historical data availability and comparability.

The project has five specific objectives:

1. **Construct a documented historical dataset** of national HPC demand and allocation using publicly available NCI, Pawsey and NCMAS sources, while retaining data provenance and reporting limitations.
2. **Improve historical comparability** by examining differences in reporting practices, computing systems, resource types and measurement units, and applying defensible standardisation or conversion methods where possible.
3. **Analyse historical demand and allocation patterns**, including requested and allocated compute, oversubscription, CPU and GPU resources, and changes in Field of Research allocation composition.
4. **Develop and evaluate forecasting approaches** appropriate for the length and quality of the available historical series, including assessments of predictive accuracy, modelling assumptions and sensitivity.
5. **Develop demand scenarios to 2030** and discuss their implications for future national research computing requirements, including uncertainty and limitations.

These objectives reflect the internship brief's emphasis on longitudinal data assembly, descriptive analysis, structural demand modelling and scenario development (Australian Academy of Science 2026).

1.4 Project Approach and Contributions

The project follows a structured analytical process covering data collection, variable definition, integration, standardisation, conversion, descriptive analysis, model development, evaluation, selection and scenario forecasting.

An important contribution is the development of a documented historical dataset that brings together information reported across different years, facilities and computing systems. The analysis also distinguishes directly reported values from derived or estimated measures, allowing the reliability and comparability of historical observations to be assessed.

Forecasting approaches are evaluated using the available annual data, and alternative scenarios are developed to examine possible changes in demand through 2030. The resulting analysis provides evidence for infrastructure planning discussions while recognising that historical allocation and requested compute do not fully capture underlying research demand.

The detailed data preparation, modelling procedures and reproducible code are documented separately in the project's technical Quarto documentation and Git repository.

Chapter 2

Data Sources and Preparation

2.1 Data Sources and Coverage

The analysis covers 13 financial years, from 2013/14 to 2025/26, represented by analytical years 2014–2026. Data were assembled from National Computational Infrastructure (NCI) annual reports, Pawsey Supercomputing Research Centre publications and National Computational Merit Allocation Scheme (NCMAS) outcomes.

The sources provide different levels of detail. Earlier publications mainly contain annual or facility-level summaries, while more recent NCMAS outcomes include project-level allocations, computing resources and research fields.

Table 2.1: Main data sources and their analytical roles.

Source	Information extracted	Use in analysis
NCI annual reports	Historical allocations, computing systems and reported resources	Historical series construction
Pawsey publications	Allocation information, computing systems and resource usage	Supplementary facility evidence and cross-checks
NCMAS outcomes	Requested and allocated resources, projects and research fields	Demand, allocation and FoR analysis
Supporting publications	Accounting rules, resource descriptions and contextual indicators	Conversion assumptions and interpretation

Research funding, workforce and research activity indicators were also examined as potential explanatory variables. Their suitability for forecasting was assessed according to historical coverage, consistency and relevance.

2.2 Data Integration and Provenance

Information extracted from the source publications was organised into two main analytical datasets: an annual demand and allocation summary, and a Field of Research (FoR) allocation composition dataset.

The integration process aligned reporting periods, identified the scope of each observation and reconciled detailed records with available annual totals. Financial years were assigned to their ending year; for example, 2013/14 was recorded as 2014.

Available provenance information was retained, including source documents, reporting periods, original values, measurement units and derivation notes.

Observations were distinguished according to whether they were directly reported, calculated from source information or estimated using documented assumptions. This distinction was preserved in the subsequent analysis.

2.3 Variable Definitions and Measurement Units

The main annual variables are requested compute, allocated compute and the difference between them. Where both measures are available on a comparable basis, the request-to-allocation ratio is calculated as:

$$R_t = \frac{D_t}{A_t}$$

where D_t represents requested compute and A_t represents allocated compute in year t.

This ratio indicates pressure on available allocations but does not measure all unmet research demand. It is calculated only when the two values have compatible units and reporting scope.

Historical records use different measurement units, including service units (SU), kilo service units (kSU), million compute hours, CPU core-hours and GPU hours. Service units are accounting measures that depend on the computing system and resource type. Consequently, values expressed in kSU are not necessarily computationally equivalent across facilities.

CPU core-hours and GPU hours were treated as separate measures. Their conversion procedures are described in the Compute Resource Conversion section.

2.4 Data Standardisation

Data standardisation included consistent year labels, research field names, computing system identifiers, resource types and provenance classifications.

Financial-year labels were aligned to their ending year, while system and resource identifiers were reviewed to distinguish CPU, GPU and aggregate observations.

Original values and reporting units were retained alongside standardised fields. Missing resource breakdowns were not treated as directly observed values, and records with different reporting scopes were identified before aggregation.

Categorical standardisation was completed separately from compute-resource conversion and FoR classification alignment, which are described in the following sections.

2.5 Compute Resource Conversion

Historical demand and allocation values were reported using different accounting conventions. To improve comparability, compute-resource conversion was undertaken according to the reporting year, computing system, resource type and available accounting documentation.

The conversion process preserved original observations and distinguished directly reported values from derived and assumed estimates. Documented system-specific factors were prioritised, while assumptions were recorded where direct conversion information was unavailable.

2.5.1 Conversion Formulas

The general relationship between service units and kilo service units is:

$$\text{SU} = \text{kSU} \times 1{,}000$$

For CPU resources reported in kSU, CPU core-hours were calculated as:

$$\text{CPU core-hours} = \frac{\text{CPU kSU} \times 1{,}000}{\text{SU per CPU core-hour}}$$

For GPU resources reported in kSU, GPU hours were calculated as:

$$\text{GPU hours} = \frac{\text{GPU kSU} \times 1{,}000}{\text{SU per GPU hour}}$$

The denominators represent the accounting factors applicable to each computing system and resource type. These factors were not assumed to be constant across all systems or reporting periods.

Where values were already reported in CPU core-hours or GPU hours, they were retained without additional service-unit conversion. Values reported in millions of hours were multiplied by 1,000,000, subject to confirmation of the original resource definition.

2.5.2 System-Specific Conversion Rules

Table Table 2.2 summarises the principal conversion rules documented in the year-by-year conversion worksheet.

Table 2.2: Main compute-resource conversion rules recorded in the year-by-year conversion worksheet.

System or resource	Reporting period	SU per resource-hour	Treatment
Raijin CPU and Magnus CPU	Selected historical years from 2013/14	1	Historical baseline; assumptions apply to some records
Gadi CPU	2020/21–2025/26	2	System-specific accounting factor
Setonix CPU	2021/22–2025/26	1	System-specific accounting factor
Setonix GPU	2021/22–2025/26	128	GPU accounting factor
Setonix Q	2025/26	256	GH200 GPU-hour conversion
Previously reported million core-hours	Selected historical years	Not applicable	Multiply by 1,000,000

The factors were applied to the corresponding system and reporting period rather than to aggregate kSU values without a resource breakdown.

For the earlier years, a factor of 1 SU per CPU core-hour was used as a baseline assumption where a more specific accounting factor was unavailable. The 2019/20 records required identification of the CPU and GPU resource components before applying separate conversions.

Historical Fornax GPU records could not be converted reliably using the available information. These observations were retained with their original units rather than assigned an unsupported conversion factor.

2.5.3 Conversion Validation and Limitations

Converted values were reviewed against the original reported units, available system-level records and published annual totals where comparable information was available.

CPU and GPU resources were kept separate because their physical computing hours are not interchangeable. Aggregate service-unit totals were not converted into CPU or GPU hours unless the relevant resource breakdown and accounting factors were available.

Missing historical GPU observations were treated as unavailable information rather than evidence of zero demand. Conversion status and supporting assumptions were retained to distinguish reported values, calculated measures and estimates.

The complete year-by-year conversion worksheet, calculations and validation code are provided in the technical project documentation.

2.6 Field of Research Classification Alignment

The FoR data were standardised in two stages to support comparisons across 2014–2026.

The first stage addressed differences between the FoR 2008-based reporting period (2014–2020) and the FoR 2020-based period (2021–2026). Selected historical fields were combined into broader reporting categories through `for_analysis_group`, while the existing standardised categories were retained for the later period.

The second stage established nine analytical groups for longitudinal comparison through `for_visual_group`. These groups were informed by the broader research-field categories used in NCI and Pawsey annual reporting. Related categories were combined where reporting detail differed across years, rather than assuming that the two FoR classification systems were directly equivalent.

Original field labels were cleaned by removing numerical prefixes, standardising case, replacing ampersands and removing excess whitespace. The cleaned labels were then assigned to the nine groups using explicit matching rules.

Table Table 2.3 summarises the analytical grouping applied to the reported research fields.

Table 2.3: Analytical FoR grouping rules used for longitudinal comparison across 2014–2026.

Analytical group	Reported fields included
Physical Sciences	Physical sciences and radioastronomy
Chemical Sciences	Chemical sciences
Earth & Environmental Sciences	Earth sciences, environmental sciences and geosciences
Engineering & Technology	Engineering and technology
Biological & Agricultural Sciences	Biological, agricultural, veterinary and food sciences
Medical & Health Sciences	Medical, health, biomedical and clinical sciences
Mathematics & Computing	Mathematical sciences, computing and information sciences
Social Sciences, Humanities & Built Environment	Psychology, economics, humanities, social sciences, design and built environment
Unclassified & Operational	Other, unreported, unclassified and Pawsey operational categories

Physical Sciences, Chemical Sciences, Earth and Environmental Sciences, and Engineering and Technology were retained as separate analytical groups. Related subfields were combined within these groups where historical reporting differed in detail.

The groupings were developed for analytical consistency and do not represent official one-to-one conversions between ANZSRC 2008 and ANZSRC 2020. Original research field labels and allocation values were preserved.

Validation checks confirmed that all 133 FoR records had the required classification and allocation information, all reported fields were assigned to one of the nine analytical groups, and annual allocation shares summed to 100%.

The complete mapping rules and validation code are available in the technical Quarto documentation.

2.7 Data Quality Assessment

Data quality was assessed using source provenance, reporting scope, measurement-unit comparability and the availability of directly observed values.

The final datasets contain both directly reported observations and reconstructed historical measures. Their reliability therefore varies across reporting years and resource types.

A specific comparability issue occurs in the 2016/17 Pawsey records, where facility-wide usage information differs in scope from NCMAS allocation data. This distinction was retained when preparing the historical FoR dataset.

Historical conversion assumptions, incomplete CPU/GPU breakdowns and differences between FoR reporting classifications introduce additional uncertainty. These issues were documented rather than removed from the analysis.

The short annual series and limited GPU coverage also constrain the forecasting methods that can be evaluated. These limitations are considered in the model evaluation and sensitivity analysis presented later in the report.

Chapter 3

Forecasting Methodology

3.1 Forecasting Framework

The forecasting analysis was designed to estimate national research supercomputing demand through 2030 using the historical data prepared in Chapter 2.

Three quantitative demand measures were examined separately: overall High-Performance Computing (HPC) demand in kilo service units (kSU), CPU demand in core-hours and GPU demand in GPU-hours. These measures have different accounting conventions and historical coverage, so they were not combined into a single physical computing measure.

A separate approach was used to examine demand across Fields of Research (FoR). Because consistent historical requests by research field were not available, FoR scenarios were developed using observed allocation shares rather than independent forecasts of researcher demand.

The analysis used relatively simple forecasting methods because the annual datasets contain few observations. Model selection considered historical predictive accuracy, the assumptions underlying future growth and the suitability of each model for projections to 2030.

3.2 Forecasting Models

Four forecasting approaches were considered: naïve forecasting, drift, linear trend and log-linear trend. The naïve and drift methods provided benchmarks against which the trend models could be evaluated.

3.2.1 Naïve and Drift Benchmarks

The naïve method assumes that the next observation will equal the latest observed value:

$$\hat{y}_{T+h} = y_T$$

where y_T is the most recent observation and h is the forecast horizon.

The drift method extends the average annual change observed between the first and last training observations:

$$\hat{y}_{T+h} = y_T + h\left(\frac{y_T - y_1}{T-1}\right)$$

These benchmarks provided a comparison with methods that explicitly estimate a trend from the full training sample.

3.2.2 Linear Trend Model

The linear trend model assumes that demand changes by a constant absolute amount each year:

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t$$

where y_t represents demand in year t, β_0 is the intercept, β_1 is the estimated annual change and ε_t is the error term.

This specification is straightforward to interpret. For example, when CPU demand is expressed in millions of core-hours, the estimated trend coefficient represents the average annual increase in that unit.

However, a linear trend assumes that the historical absolute change continues over the forecast period. It does not account directly for future changes in computing technology, allocation policies or research funding.

3.2.3 Log-Linear Trend Model

The log-linear model assumes approximately constant proportional growth:

$$\log(y_t) = \beta_0 + \beta_1 t + \varepsilon_t$$

The corresponding growth rate is:

$$g = (e^{\beta_1} - 1) \times 100\%$$

where g is the implied annual percentage growth rate.

Unlike the linear model, the log-linear specification allows the absolute annual increase to grow as demand rises. This can produce substantially higher long-term projections when historical growth is strong.

The model requires positive demand observations and assumes that proportional growth remains relevant over the forecast horizon. Its projections were therefore considered alongside the linear model rather than accepted solely on the basis of historical fit.

3.3 Forecasting Targets and Historical Coverage

The forecasting models were applied separately to the available demand measures.

Table 3.1: Forecasting targets and historical coverage.

Forecasting target	Historical period	Measurement unit	Main consideration
Overall HPC demand	2014–2026	kSU	Historical reporting and accounting comparability
CPU demand	2014–2026	Million CPU core-hours	System-specific conversion assumptions
GPU demand	2022–2026	Million GPU-hours	Shorter series and estimated resource-level values

The overall HPC and CPU series cover 13 annual observations. GPU demand required a shorter modelling period because earlier GPU values were constructed using hypothetical conversion scenarios rather than sufficiently supported resource-level demand observations.

The GPU analysis therefore focused on 2022–2026. These recent values also include derived estimates and working conversion assumptions, which were considered when interpreting forecast reliability.

All models were fitted separately for each demand measure. Forecasts were expressed in the corresponding units rather than combining CPU core-hours and GPU hours.

3.4 Model Evaluation

3.4.1 Rolling-Origin Cross-Validation

Forecasting performance was assessed using expanding-window, one-step-ahead rolling-origin cross-validation.

Under this approach, a model was fitted using the observations available up to a particular year and then used to predict the following year. The training window was expanded by one observation for each subsequent evaluation.

This approach preserves the chronological order of the data and avoids using future observations to estimate earlier forecasts.

For overall HPC and CPU demand, the initial training period covered 2014–2020. One-step-ahead forecasts were evaluated for 2021–2026, producing six forecast errors for each model.

For GPU demand, the initial training period covered 2022–2024, with forecasts evaluated for 2025 and 2026. This provided only two forecast errors, making GPU model comparisons less reliable than those for overall HPC and CPU demand.

3.4.2 Forecast Accuracy Measures

Model accuracy was assessed using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

RMSE was calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

MAE was calculated as:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

where y_t is the observed value, $\hat{y}_t$ is the forecast value and n is the number of evaluated forecasts.

Both measures are expressed in the same units as the forecasting target. Lower values indicate better predictive accuracy.

RMSE gives greater weight to larger errors, while MAE measures the average absolute error. Both were considered because individual observations can have a substantial influence on accuracy measures when the evaluation period is short.

Model performance was compared within each demand measure. RMSE and MAE values were not compared directly across HPC, CPU and GPU series because their measurement units differ.

3.5 Model Selection and Sensitivity Analysis

Model selection considered predictive accuracy, interpretability and the assumptions required to extend historical trends through 2030.

A model with the lowest historical error was not automatically selected if its growth assumptions produced projections that were difficult to justify from the available evidence.

The linear and log-linear specifications were compared particularly closely because they represent different future growth patterns. Linear models assume a constant absolute annual increase, while log-linear models imply compounded proportional growth.

A sensitivity analysis was also conducted by fitting trend models with and without the 2026 observation. This examined how strongly the latest reported year influenced estimated growth and the resulting projections.

The comparison was intended to assess sensitivity to the historical sample rather than determine whether the 2026 observation represented a temporary or permanent change.

Forecast trajectories derived from different model specifications were treated as alternative assumptions, not formal statistical prediction intervals.

3.6 Field of Research Scenario Development

FoR scenarios were developed separately from the overall HPC, CPU and GPU forecasting models.

The available FoR data describe the distribution of allocated computing resources across research fields. They do not provide a consistent historical series of independently observed demand for every research field.

A top-down approach was therefore used. The selected overall HPC demand projection was distributed across the nine harmonised FoR groups using alternative assumptions about future allocation shares.

For research field i and forecast year t, the indicative allocation-share-based demand was calculated as:

$$\hat{D}_{i,t} = \hat{D}_t \times s_i$$

where $\hat{D}_t$ is the overall forecast HPC demand and s_i is the assumed share for research field i.

Three historical periods were examined to construct alternative share assumptions:

Table 3.2: Historical periods used to construct FoR allocation-share scenarios.

Scenario	Historical basis	Purpose
Recent composition	2021–2026	Reflect the more recent FoR reporting period
Full-period composition	2014–2026	Reflect the broader historical allocation pattern
Earlier composition	2014–2020	Examine sensitivity to the earlier research-field distribution

For each scenario, the relevant historical allocation shares were used to derive a distribution across the harmonised research-field groups.

Differences in category coverage were considered when preparing the historical averages. Missing research-field observations were not automatically treated as zero allocations, and scenario shares were normalised where required to produce a complete distribution.

These scenarios illustrate how alternative assumptions about research-field composition affect the distribution of forecast demand. They do not represent independently validated forecasts of future requests by FoR.

3.7 Forecast Horizon and Interpretation

The final forecasting horizon covers 2027–2030.

After model evaluation and selection, the selected specifications were fitted to the available historical data to generate future projections. Alternative growth assumptions and sensitivity specifications were retained to examine how the forecasts changed under different modelling choices.

The resulting projections represent extrapolations of historical demand measures under stated assumptions. They are not direct estimates of future computing capacity requirements, and they do not account explicitly for changes in national research policy, future computing architectures or potential constraints on access to HPC resources.

The limited number of annual observations, differences in historical accounting and the short GPU series restrict the certainty of the projections. These limitations were considered when comparing model results and interpreting the 2030 scenarios.

Chapter 4

Results

4.1 Initial Analysis

The initial analysis examined historical HPC demand and allocation, CPU and GPU resources, and the distribution of allocations across Fields of Research (FoR). The results were used to identify trends, assess differences in reporting coverage and determine which series were suitable for forecasting.

4.1.1 Historical HPC Demand and Allocation

Figure Figure 4.1 shows historical demand and allocation between 2014 and 2026.

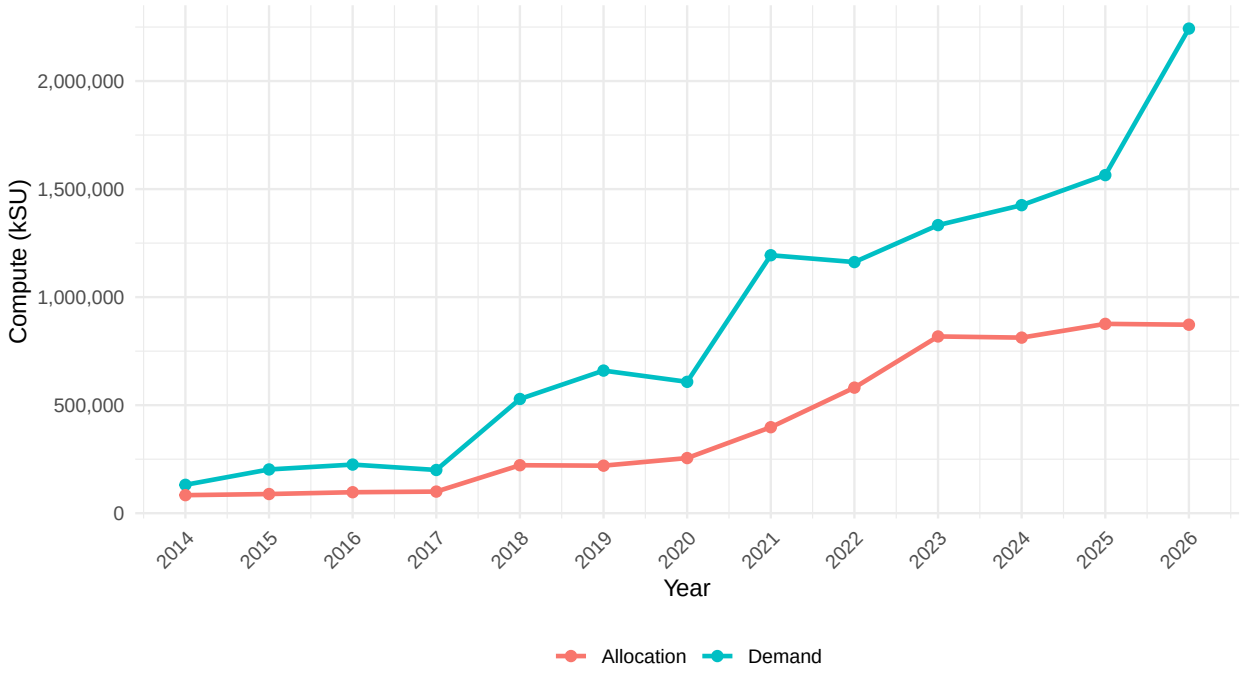


Figure 4.1: Historical HPC demand and allocation, 2014–2026.

Demand increased from approximately 131,356 kSU in 2014 to 2,242,817 kSU in 2026, while allocation increased from approximately 83,505 kSU to 872,288 kSU.

Demand exceeded allocation throughout the analytical period. The largest recorded difference occurred in 2026, when requested resources exceeded allocations by approximately 1.37 million kSU. The fulfilment rate declined from 56.0% in 2025 to 38.9% in 2026.

The series shows changes in growth around 2018, 2021 and 2026. However, historical reporting differences and conversion assumptions affect comparability. The increase in recorded demand cannot be attributed entirely to changes in researcher requirements.

4.1.2 CPU Demand and Allocation

CPU resources were analysed separately from aggregate HPC demand because service-unit accounting differs across computing systems.

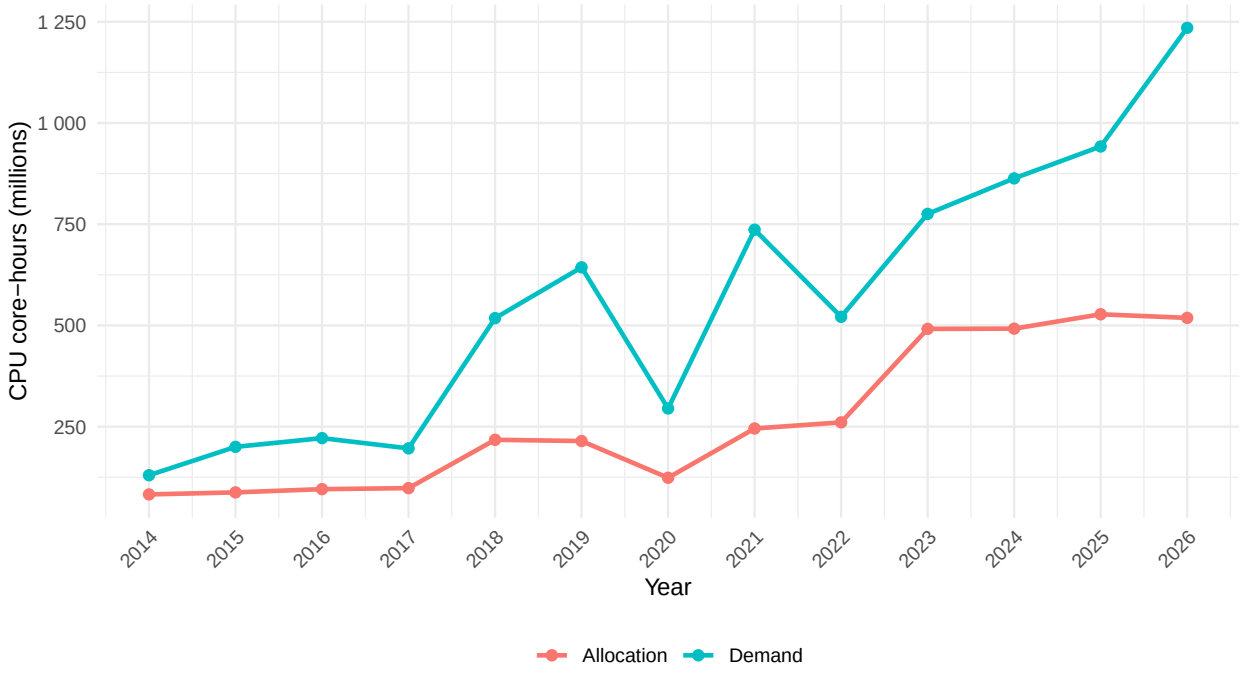


Figure 4.2: Historical CPU demand and allocation, 2014–2026. Earlier values include estimates.

Figure Figure 4.2 shows the development of CPU demand and allocation in core-hours. The series provides a separate measure of CPU resource requirements after applying the conversion rules documented in Chapter 2.

The historical values include estimated observations, particularly in earlier years. Consequently, the long-term trend is useful for examining the overall direction of CPU requirements, but individual year-to-year changes should not be interpreted as exact changes in physical resource consumption.

4.1.3 GPU Demand and Allocation

GPU demand was examined separately because the historical records have more limited coverage and require additional conversion assumptions.

Figure Figure 4.3 retains the baseline series and historical low–high conversion scenarios from the technical analysis.

The historical GPU series is less certain than the CPU series. The 2014–2021 values include hypothetical conversion scenarios, while the 2022–2026 period contains a mixture of reported and derived resource estimates.

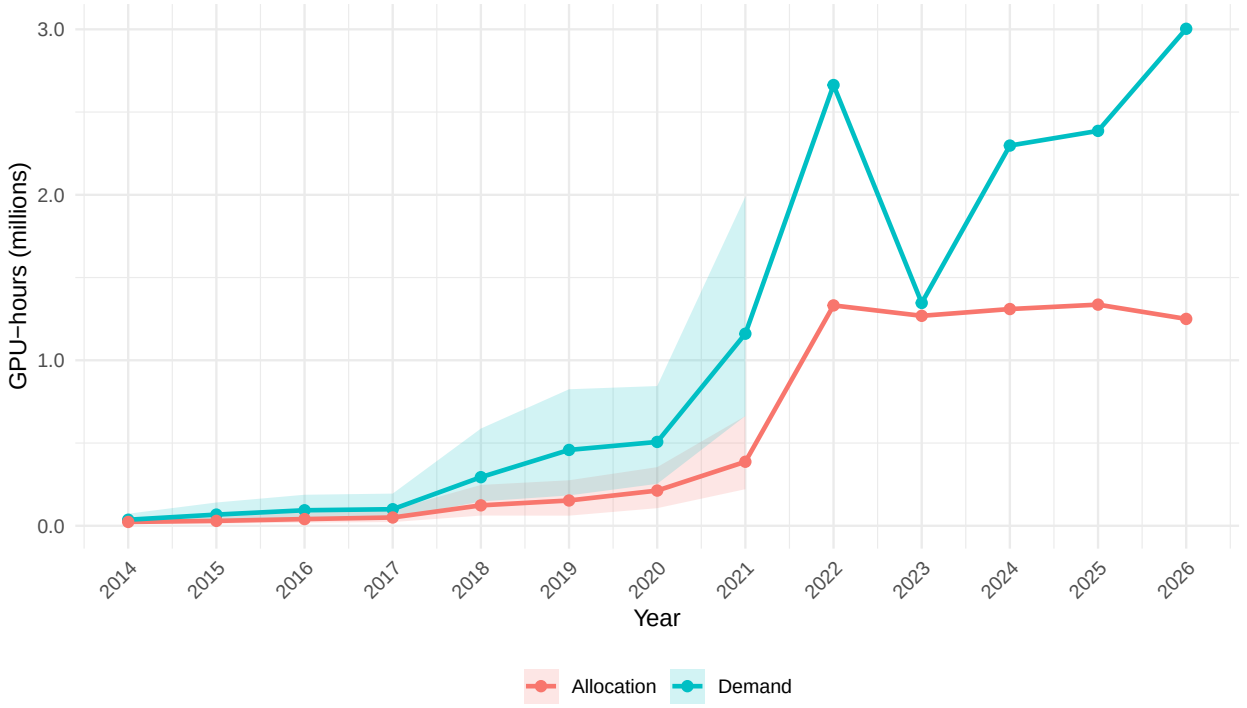


Figure 4.3: Historical GPU demand and allocation, 2014–2026. Shaded areas show historical conversion scenarios, not statistical confidence intervals.

The shaded ranges represent uncertainty associated with historical conversion assumptions rather than statistical prediction intervals.

For this reason, the forecasting analysis uses the shorter 2022–2026 GPU series. This reduces reliance on earlier conversion assumptions but leaves only five annual observations for model development.

4.1.4 Field of Research Allocation Composition

The FoR analysis examines how HPC allocations were distributed across research disciplines.

To support longitudinal comparison, the reported research fields were consolidated into nine analytical groups. These groupings were developed for the project and do not represent official one-to-one conversions between FoR 2008 and FoR 2020.

Across 2014–2026, Engineering and Technology accounted for an average annual allocation share of approximately 25.2%, followed by Earth and Environmental Sciences at 23.5%, Physical Sciences at 19.3% and Chemical Sciences at 18.5%.

The composition differs between reporting periods. During 2021–2026, Engineering and Technology accounted for approximately 34.9% of average annual allocations, while Physical Sciences represented 22.1%, Chemical Sciences 17.8% and Earth and Environmental Sciences 14.0%.

These results are based on the average of annual allocation shares rather than pooled allocations across all years. Differences between periods must also be interpreted in the context of changes in classification, reporting coverage and the 2016/17 Pawsey-wide records.

The FoR results describe how resources were allocated, not independently observed demand from individual research fields.

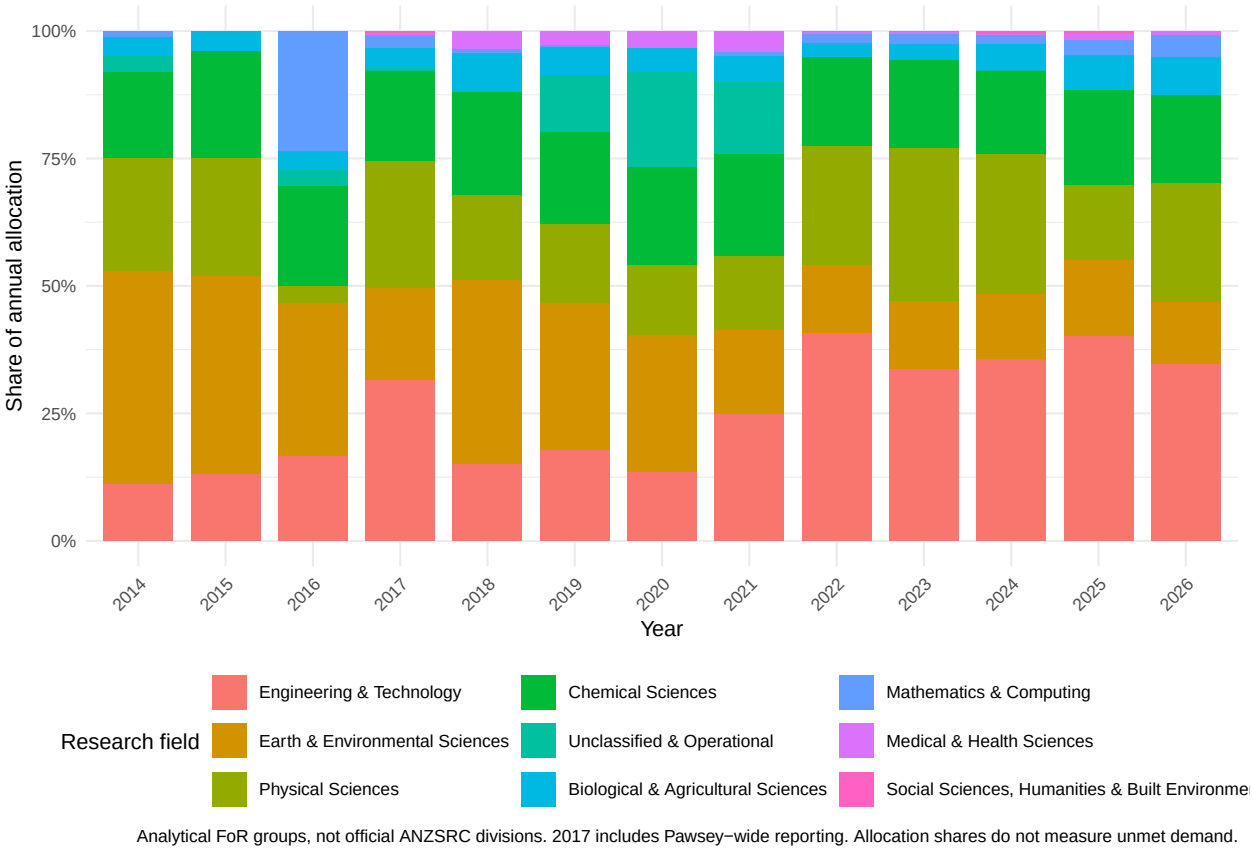


Figure 4.4: Annual HPC allocation shares by harmonised Field of Research group, 2014–2026.

4.2 Forecast Model Evaluation

The forecasting models were evaluated using rolling-origin, one-step-ahead predictions. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were used to compare forecast accuracy.

4.2.1 Overall HPC Demand

The linear trend achieved the lowest RMSE and MAE for overall HPC demand during the 2021–2026 evaluation period.

Its RMSE was approximately 290,898 kSU, compared with 314,588 kSU for the drift model, 378,864 kSU for the naïve benchmark and 511,614 kSU for the log-linear trend.

The linear trend was therefore selected as the central model. However, the results are based on six evaluation years, and the substantial increase in 2026 affects the fitted trend.

4.2.2 CPU Demand

For CPU demand, the linear and log-linear models performed similarly.

The linear trend achieved the lowest RMSE of approximately 157.72 million core-hours, while the log-linear model achieved a slightly lower MAE.

The linear trend was preferred because it provided competitive forecast accuracy without assuming continued proportional growth. The log-linear model was retained as an alternative growth scenario.

4.2.3 GPU Demand

GPU model evaluation was limited to forecasts for 2025 and 2026.

The naïve benchmark achieved the lowest RMSE, approximately 0.441 million GPU-hours. However, two evaluation years provide insufficient evidence to establish a reliable long-term forecasting model.

The naïve model was retained as a reference, while linear and log-linear trends were used to explore alternative growth assumptions.

4.3 Demand Forecasts to 2030

4.3.1 Overall HPC Demand

The selected linear trend projects overall HPC demand of approximately 2.02 million kSU in 2027, increasing to 2.50 million kSU in 2030.

The model excluding the 2026 observation produces a lower 2030 projection of approximately 2.29 million kSU. In contrast, the log-linear model projects approximately 6.48 million kSU.

The difference illustrates the sensitivity of longer-term projections to the latest observation and the assumed growth pattern. The log-linear trajectory should not be interpreted as a validated upper bound because it performed less well during historical evaluation.

4.3.2 CPU Demand

The central CPU forecast increases from approximately 1.13 billion core-hours in 2027 to 1.37 billion core-hours in 2030.

Excluding 2026 reduces the 2030 projection to approximately 1.26 billion core-hours, while the log-linear model produces approximately 2.48 billion core-hours.

The central linear model provides a reference for planning, while the alternative trajectories illustrate the consequences of different growth assumptions.

4.3.3 GPU Demand

GPU projections show greater uncertainty because of the shorter historical series.

The naïve benchmark maintains the latest estimate at approximately 3.00 million GPU-hours through 2030. The linear trend projects approximately 3.37 million GPU-hours, while the log-linear trend projects approximately 3.68 million GPU-hours.

These are exploratory scenarios rather than reliable long-term predictions. The short historical record limits the ability to determine whether GPU demand will stabilise or continue growing.

4.4 Field of Research Scenarios

The FoR scenarios apply alternative historical allocation compositions to the projected overall HPC demand.

Three reference periods were considered: the earlier period (2014–2020), the recent period (2021–2026) and the full period (2014–2026).

The recent-period composition gives greater weight to Engineering and Technology, while the earlier-period composition places greater emphasis on Earth and Environmental Sciences.

These differences affect the indicative distribution of future HPC resources across disciplines. However, the estimates are conditional on historical allocation shares and do not represent independently modelled FoR-specific demand.

The comparison is therefore more useful for examining the sensitivity of future resource distribution to allocation assumptions than for predicting precise disciplinary requirements.

4.5 Summary of Results

The historical analysis identifies increasing pressure between recorded HPC demand and available allocations, particularly in 2026. CPU and GPU demand also show different measurement and forecasting challenges, with GPU analysis constrained by the shorter and less consistent historical record.

The linear trend provides the central forecast for overall HPC and CPU demand. By 2030, these models project approximately 2.50 million kSU and 1.37 billion CPU core-hours, respectively.

GPU forecasts remain exploratory, while FoR scenarios show how assumptions about historical allocation composition affect the indicative distribution of future resources.

These results provide the basis for the discussion of forecasting uncertainty and national HPC planning implications in the following chapter.

Chapter 5

Discussion

5.1 Interpretation of the Main Findings

The historical records show a persistent difference between requested and allocated HPC resources. This difference was particularly pronounced in 2026. It indicates pressure within the reported allocation system, although it cannot be interpreted as a direct measure of all unmet research demand. Researchers may adjust requests to expected capacity, and some potential demand is not captured by the allocation process.

The forecast evaluation favoured a linear trend for overall HPC demand and CPU demand. These models produced competitive one-step-ahead accuracy and avoided the assumption that historical proportional growth would continue unchanged through 2030. Nevertheless, the central projections remain conditional on the historical accounting measures and the assumption that recent trends provide useful information about future requests.

The GPU results require greater caution. The model comparison used only two evaluation years, and the 2022–2026 modelling series includes derived estimates. Differences between the naïve, linear and log-linear projections should therefore be read as exploratory outcomes under different assumptions, not as evidence that one growth path will occur.

5.2 Implications for National HPC Planning

The analysis provides a documented basis for discussing the scale and direction of reported HPC demand. The projected increase in overall requests suggests that national infrastructure planning should consider demand growth alongside the historical difference between requests and allocations.

However, forecast kSU values are accounting measures rather than direct measures of interchangeable computing capacity. Hardware changes, system-specific service-unit factors, GPU adoption and the design of allocation schemes may all affect the relationship between reported requests and the resources required to meet them.

The FoR scenarios provide a complementary view of possible distributional changes. They show that the implied allocation of future resources across disciplines depends on which historical period is used to represent research-field composition. These results are suitable for sensitivity analysis but should not be treated as direct forecasts of unmet demand within individual disciplines.

5.3 Limitations and Future Work

The principal limitation is the short and uneven historical record. Overall HPC and CPU demand contain 13 annual observations, while the GPU modelling period contains only five. Historical differences in reporting

scope and conversion assumptions also limit comparisons across facilities and years.

The analysis relies on publicly available records. It does not incorporate confidential application-level information, direct measures of rejected or discouraged requests, or future changes in policy and computing technology. Consequently, the forecasts should be updated as more consistent annual records become available.

Future work could improve the analysis by obtaining longer and more detailed requested-versus-allocated series, validating conversion factors directly with facility operators, and recording CPU and GPU demand using stable resource definitions. Consistent FoR-level request data would also allow research-field demand to be modelled directly rather than approximated through allocation shares.

Chapter 6

Conclusion

This project assembled and standardised publicly available Australian national research HPC records covering 2014–2026 and developed demand projections through 2030. It combined data preparation, conversion validation, descriptive analysis, rolling-origin forecast evaluation and scenario development.

The historical analysis found that recorded HPC requests exceeded allocations throughout the study period. For overall HPC demand and CPU demand, the linear trend was selected as the central forecasting approach. The corresponding 2030 projections were approximately 2.50 million kSU and 1.37 billion CPU core-hours. Alternative models produced substantially different outcomes, particularly under compounded proportional growth assumptions.

GPU forecasts and FoR scenarios were treated more cautiously because of limited historical coverage and the distinction between allocated resources and independently observed demand. The results offer a transparent reference for future infrastructure discussions rather than a precise estimate of unconstrained national computing requirements.

The main contribution is a reproducible analytical framework that makes the underlying data sources, conversion assumptions, model choices and forecasting limitations explicit. The accompanying technical Quarto documentation and Git repository preserve the detailed implementation for review and future updates.

Chapter 7

References

Australian Academy of Science. 2024. "The Future Computing Needs of the Australian Science Sector." https://www.science.org.au/.

Australian Academy of Science. 2026. *Internship Project Brief: Forecasting Demand for National Research Supercomputing.*

National Computational Infrastructure. n.d. "Who We Are." Accessed October 4, 2026. https://nci.org.au/.

Pawsey Supercomputing Research Centre. n.d. "About Pawsey." Accessed October 4, 2026. https://pawsey.org.au/.